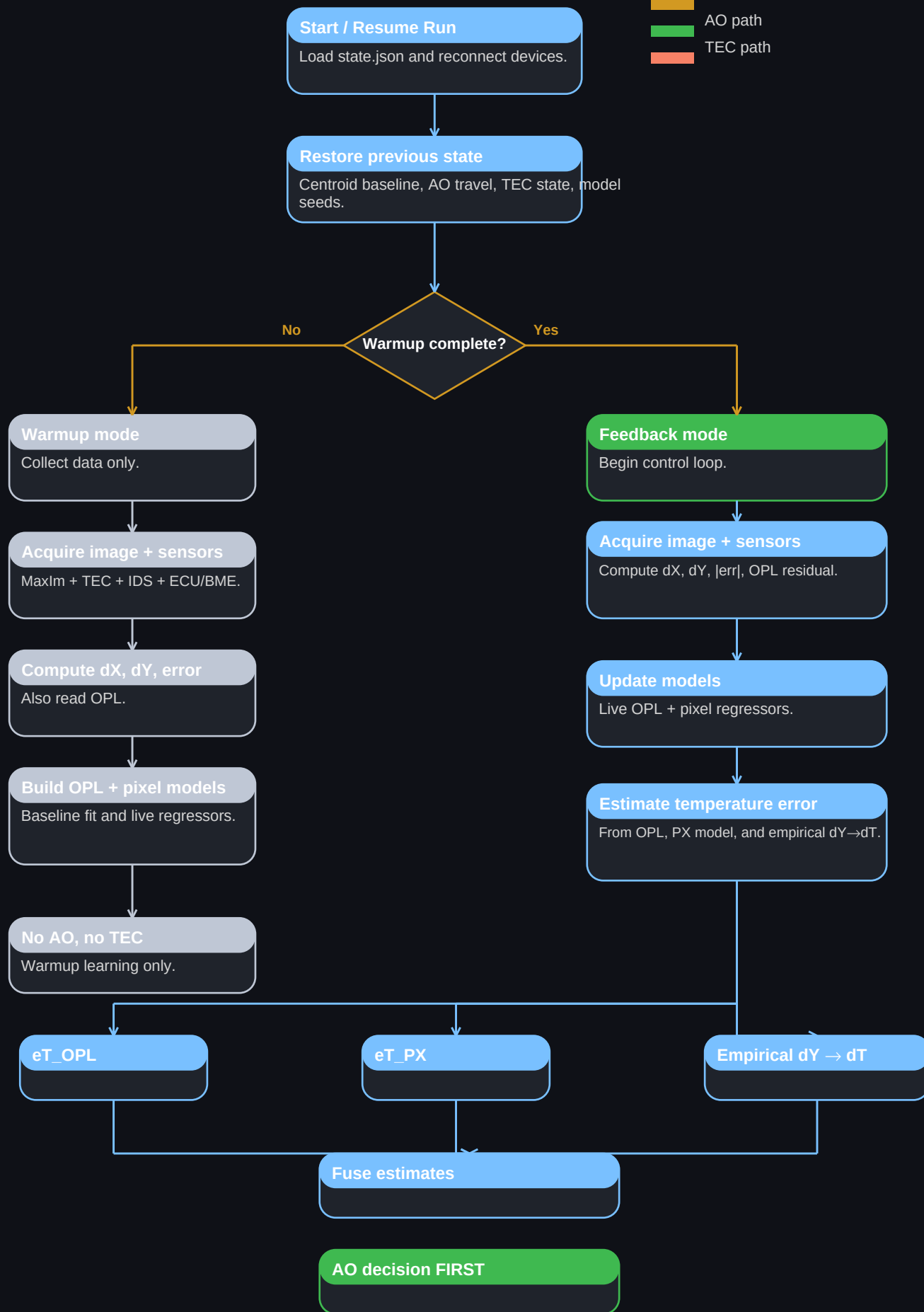
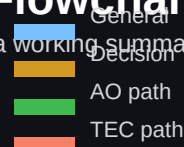


Stage-2 Hybrid Controller - Mermaid Style Flowchart

Readable multi-page overview of AO-priority logic with temperature backup. This is a working summary and is updated as the controller



Continues on next page → AO gating and AO action logic.

Stage-2 Hybrid Controller - AO Logic

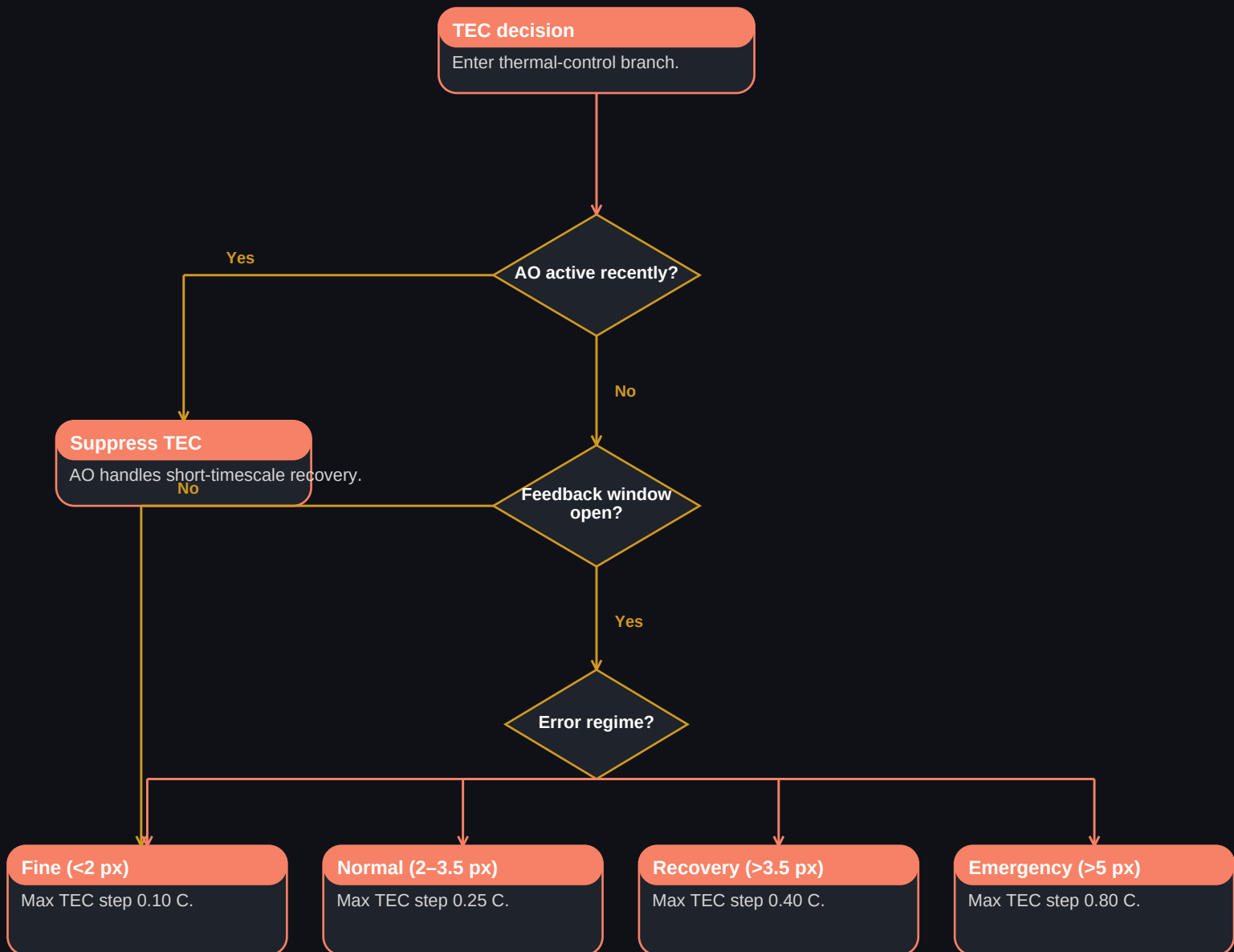
AO acts first for fast centroid recovery. Thresholds and limits shown here.



Continues on next page → TEC gating and regime selection.

Stage-2 Hybrid Controller - TEC Gating and Regimes

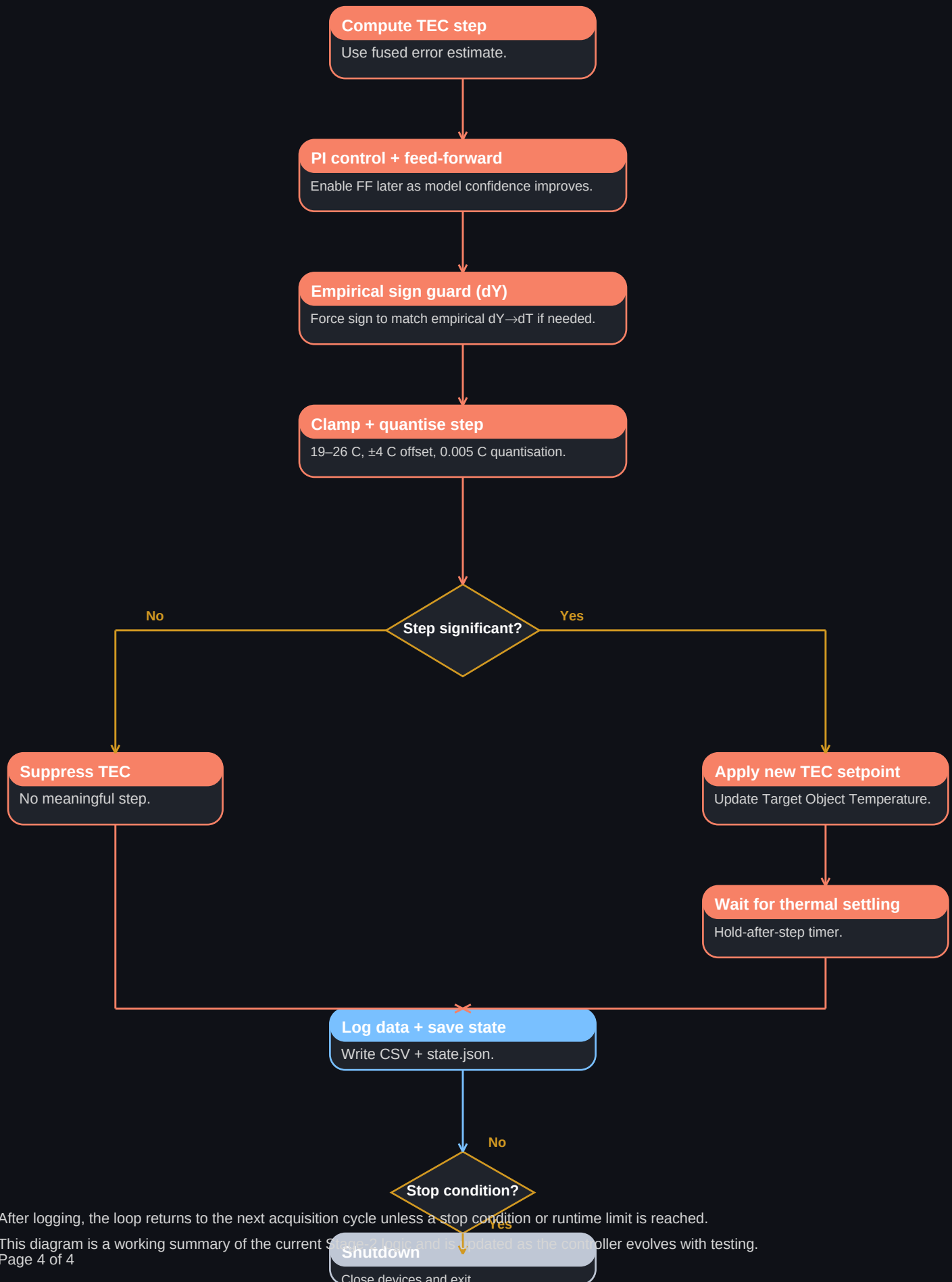
TEC acts second and is slower. It is gated while AO is actively correcting.



Continues on next page → TEC command generation, step significance, and shutdown.

Stage-2 Hybrid Controller - TEC Command and Loop Finish

Final thermal-command stage, safety checks, and loop closure.



After logging, the loop returns to the next acquisition cycle unless a stop condition or runtime limit is reached.

This diagram is a working summary of the current Stage-2 logic and is updated as the controller evolves with testing.